

DSAA3071 Week 5 Candidate-v3.2 Development Note

Development split only. held-out not run; not a final accuracy claim.

Technical summary. C32 is the strongest development condition so far on aggregate MAE: question MAE improves from R1=2.614 to C32=2.557, and total MAE improves from R1=20.143 to C32=14.429. The result is promising but not final: Q8 remains the main unresolved error source, Q6 worsens versus R1, and C32 still has slightly higher severe-error rate than R1.

What Was Tested

1. Same data
Seven anonymous development students from the reviewed DSAA3071 Week 5 transcript snapshot.

2. Same gold
Official per-question scores from `primary_scores.csv`; no official rationales are available.

3. New package
C32 uses `candidate-v3.2` plus `rubric_v2.json` to reduce over-harsh grading on Q7-Q9.

This note covers DSAA3071 / week5_test on branch `codex/dsaa3071-week5-dev-transcripts` at analysis commit `7f4e06a`.

Reproducibility Anchors

Anchor	Value
Metric record	<code>experiments/records/DSAA3071-week5-candidate-v31-dev-plan/dev-metrics-deepseek-c32.json</code>
Run protocol	<code>experiments/records/DSAA3071-week5-candidate-v31-dev-plan/RUN-PROT0COL-C32.md</code>
Prompt snapshot	<code>experiments/records/DSAA3071-week5-candidate-v31-dev-plan/prompts/grade_candidate_v3_2_strict_schema.txt</code>
Rubric	<code>experiments/records/DSAA3071-week5-prep/rubric_v2.json</code>
Skill snapshot	<code>experiments/skill_versions/skill_candidate_v3_2.json</code>
Data snapshot	<code>95e744f5811d...</code>
Text source hash	<code>9f9be72bf088...</code>

Prompt And Skill Evolution

Why this matters. The DSAA3071 experiment is not only a model benchmark. It is a grading-skill development experiment for concept-heavy short-answer and proof questions. The conditions below show how the prompt/skill moved from standard-answer matching toward evidence-first, type-aware, and open-ended grading.

Cond.	Prompt / skill package	Main grading behavior	Why it was tested
B0	baseline prompt + rubric_v0.json	Simple frozen-rubric grading from the extracted official solution. It mainly checks whether the answer matches the expected solution and point criteria.	Reference starting point. Shows how the uncalibrated framework behaves on DSAA3071.
R1	same baseline prompt + rubric_v1.json	Keeps the prompt constant but upgrades the rubric into concept/key-term elements, score bands, full-credit rules, and material-error caps.	Isolates whether a better rubric helps before changing the grading skill.
C3	candidate-v3 prompt + rubric_v1.json	Adds a type-aware grading algorithm: multiple choice, short answer, calculation, algorithm, proof, and essay are scored by different rules. It uses evidence before score and separates key-term, concept, and relation evidence.	Tests whether the grading skill itself improves over the baseline prompt under the same rubric.
C31-r2	candidate-v3.1 open-ended prompt + rubric_v1.json	Adds open-ended adequacy: the standard answer is an anchor, not an exhaustive whitelist. Valid, relevant, non-contradictory constructions or explanations can receive credit even when phrased differently.	Addresses over-harsh scoring on open-ended answers, especially Q8 and Q9.
C32	candidate-v3.2 prompt + rubric_v2.json	Adds official-style adequacy and “avoid being overly harsh” rules. It also adds targeted policies for Q7 proof locality, Q8 power-of-two enumerators, and Q9 Church-Turing conceptual evidence.	Current calibrated package. Tests whether targeted prompt plus targeted rubric reduce aggregate error.

What changed from C3 to C31-r2 C31-r2 keeps the core candidate-v3 algorithm but adds an explicit open-ended adequacy rule. This tells the grader not to reject a valid answer merely because it is not listed in the official solution.

What changed from C31-r2 to C32 C32 adds official-style tolerance plus question-specific policies. It is less ideal-answer-completeness driven, but still requires the answer to satisfy the actual task and avoids giving credit for material contradictions.

Important experimental interpretation. R1 is close to a rubric-only comparison against B0. C3 is closer to a skill-prompt comparison against R1. C32 is a calibrated package-level comparison because both the prompt and rubric changed. Therefore C32 should be read as the best current grading package on dev, not as a pure one-factor ablation.

Rubric Evolution

Rubric	Used by	Design	Main limitation / improvement
rubric_v0	B0	Draft rubric extracted from the solution PDF. Most questions are represented as point criteria tied closely to the expected answer.	Good enough to start the benchmark, but weak for concept answers because it gives little guidance about semantic equivalents, partial understanding, or local mistakes.
rubric_v1	R1, C3, C31-r2	Concept/key-term rubric. Adds scoring elements, evidence levels, score bands, full-credit rules, and material-error caps. It distinguishes absent, mentioned-only, partial understanding, demonstrated, and misused evidence.	Improves structure and partial-credit behavior, but Q8/Q9 still showed over-harsh or unstable behavior in dev.
rubric_v2	C32	Targeted calibration from C31-r2 error diagnosis. Adds official-style tolerance and specific rules for Q7 proof locality, Q8 $\theta^*(2^n)$ enumerator behavior, and Q9 broad Church-Turing evidence.	Best aggregate dev MAE so far, but Q8 remains unresolved and Q6 worsens versus R1.

Q7 proof locality	Q8 enumerator policy	Q9 conceptual essay policy
Preserve credit for each correctly demonstrated proof direction. A local nonmembership/rejection mistake should not erase unrelated construction credit.	The target language is $\theta^*(2^n)$. Distinguish powers of two from linear even lengths $2n$, invalid extra outputs, wrong base cases, and vague loop mechanisms.	Accept broad valid evidence for the Church-Turing thesis when it supports effective computability. Do not require exact labels such as lambda calculus if the explanation is equivalent and non-contradictory.

Data discipline. These rubric changes were calibrated from aggregate development errors and official solutions. The report does not copy raw student answer text, and official scores provide per-question gold scores but not detailed official rationales.

Key Findings With Visual Evidence

Best question MAE

2.557

C32, lower is better

Best total MAE

14.429

C32, lower is better

Exact agreement

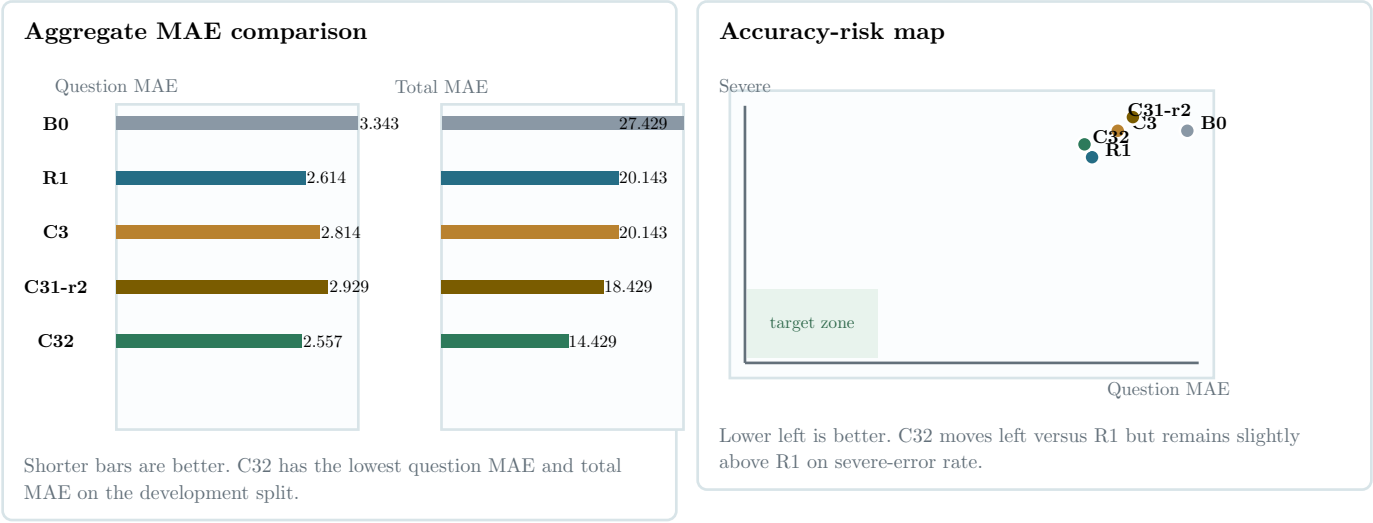
52.9%

C32 ties C31-r2

Severe error

22.9%

C32, still above R1



Condition Details

Cond.	Description	Validation	Tokens	Q MAE	Total MAE	Severe
B0	baseline prompt + rubric v0	passed (7/7)	63448	3.343	27.429	24.3%
R1	baseline prompt + rubric v1	passed (7/7)	92633	2.614	20.143	21.4%
C3	candidate-v3 prompt + rubric v1	passed (7/7)	104803	2.814	20.143	24.3%
C31-r2	candidate-v3.1 r2 open-ended adequacy prompt + rubric v1	passed_after_recovery (7/7)	107658	2.929	18.429	25.7%
C32	candidate-v3.2 prompt + rubric v2	passed (7/7)	116758	2.557	14.429	22.9%

C32 improves aggregate MAE while preserving the same seven-student development split. This table should be read as calibration evidence, not as a final leaderboard.

Per-Question Error Concentration

Question	B0	R1	C3	C31-r2	C32	Best
Q5	7.000	4.857	5.857	2.571	2.714	C31-r2
Q6	2.143	1.714	1.857	2.714	4.429	R1
Q7	4.429	2.429	4.429	5.000	2.429	R1, C32
Q8	3.857	5.714	4.857	5.000	5.571	B0
Q9	14.857	9.714	9.714	11.714	9.000	C32
Q10	1.143	1.714	1.429	2.286	1.429	B0

The strongest C32 improvements are Q7 and Q9 relative to C31-r2. The biggest blocker is still Q8: C32 Q8 MAE is 5.571, almost unchanged from R1 and worse than B0.



C32 Q7-Q9 Development Error Table

Student	Question	Gold	C32	Error
S017	Q7	20.000	17.000	−3.000
S017	Q8	9.000	5.000	−4.000
S017	Q9	25.000	12.000	−13.000
S021	Q7	15.000	16.000	1.000
S021	Q8	0.000	5.000	5.000
S021	Q9	12.000	10.000	−2.000
S002	Q7	10.000	15.000	5.000
S002	Q8	10.000	3.000	−7.000
S002	Q9	25.000	25.000	0.000
S015	Q7	10.000	13.000	3.000
S015	Q8	10.000	4.000	−6.000
S015	Q9	25.000	17.000	−8.000
S020	Q7	18.000	18.000	0.000
S020	Q8	10.000	1.000	−9.000
S020	Q9	10.000	2.000	−8.000
S016	Q7	0.000	0.000	0.000
S016	Q8	8.000	5.000	−3.000
S016	Q9	25.000	5.000	−20.000
S022	Q7	10.000	5.000	−5.000
S022	Q8	10.000	5.000	−5.000
S022	Q9	25.000	13.000	−12.000

This table uses anonymous IDs only and does not copy student answer text. It shows why the result is still mixed: Q7 is mostly controlled, Q9 improved but remains under-scored for several students, and Q8 remains unstable.

Metric Definitions And Method

- `question_score_mae`: mean absolute error over all student-question pairs. Lower is better.
- `normalized_question_score_mae`: mean absolute error divided by question max score. Lower is better.
- `question_exact_agreement`: fraction of student-question pairs where model score exactly equals official score. Higher is better.
- `severe_error_rate_abs_ge_5`: fraction of student-question pairs with absolute error at least 5 points. Lower is better.
- `total_score_mae`: mean absolute error between model total and official total per student. Lower is better.

The model run used DeepSeek public API, `deepseek-v4-pro`, text-only inputs, the reviewed transcript snapshot, and packet `C32-dev-reviewed-r1`. The model output schema validation passed for all seven development students.

Limitations And Next Step

- held-out not run; this is not a final accuracy claim.
- The development split has only seven anonymous students.
- Official scores provide per-question numbers but not detailed official rationales.
- C32 changes rubric, prompt, and skill together by design, so it evaluates the calibrated package rather than one isolated factor.
- Next step: diagnose Q8 specifically before freezing the skill or running a held-out test.